

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
222TCS002	DEEP LEARNING	PROGRAM CORE 3	3	0	0	3

Preamble: This course introduces the core concepts of deep learning and also provides an insight into recent developments in the field. The concepts covered in the course include Neural Network Optimization techniques, Regularization, Convolutional Neural networks, Recurrent Neural Networks, Word Embedding and Transformers. This course helps the students to develop solutions to real world applications using deep learning techniques.

Course Outcomes: After the completion of the course the student will be able to:*

CO 1	Analyse the challenges in learning of Neural Networks and develop solutions to overcome the issues (Cognitive knowledge level: Apply)
CO 2	Construct convolutional neural networks for deep learning applications (Cognitive knowledge level: Apply)
CO 3	Make use of recurrent neural network and its variants in relevant application areas (Cognitive knowledge level: Analyze)
CO 4	Apply the deep learning techniques in natural language based applications (Cognitive knowledge level: Analyze)
CO 5	Distinguish the transformer architecture from earlier architectures (Cognitive knowledge level: Analyze)
CO 6	Design, develop and implement solutions based on Deep Learning concepts and techniques (Cognitive knowledge level: Create)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1:An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2:An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3:An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4:An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5:An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6:An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7:An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							
CO6							

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	40
Analyse	20
Evaluate	Can be evaluated using miniprojects/assignments
Create	Can be evaluated using miniprojects/assignments

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation : 40 marks

Micro project/Course based project : 20 marks

Course based task/Seminar/Quiz : 10 marks

Test paper, 1 no. : 10 marks

The project shall be done individually. Group projects not permitted.

Test paper shall include minimum 80% of the syllabus.

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the University. There will be two parts; Part A and Part B. Part A contain 5 numerical questions with 1 question from each module, having 5 marks for each question. (such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students). Students shall answer all questions.

Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

Total duration of the examination will be 150 minutes.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Apply the Adam optimization algorithm to an example of your choice and implement it using appropriate tools..
2. Explore the validity of the statement: “Computational complexity due to Batch normalization is proportional to the size of the mini batch”.

Course Outcome 2 (CO2)

1. Design and sketch a CNN for any application of your choice.
2. Differentiate between a CNN and an autoencoder in terms of architecture and applications.
3. Demonstrate the use of transfer learning in natural language processing.

Course Outcome 3(CO3):

1. Illustrate the workflow inside an RNN by using an example.
2. In comparison to an RNN, how many orders of excess computation are there in an LSTM?
3. Implement any sequence to sequence based application using RNN and LSTM. Analyse which one of these two is more relevant for your application.

Course Outcome 4 (CO4):

1. Provide schematic diagrams showing input, output and hidden layers along with their connectivity for the word2vec embedding method.
2. Analyze how good BERT is in the application area of automated question answering.
3. Derive the loss functions used for training of a GAN.

Course Outcome 5 (CO5):

1. Differentiate between self-attention and multi-head attention.
2. Justify the statement: "Vision transformer is an adaptation of the transformer architecture for computer vision applications' '.
3. Analyze the computational complexities involved in large scale pretraining of transformers.

Course Outcome 6 (CO6):

1. Develop a deep learning model as a solution to a real world problem and analyse its performance.

Estd.



2014

Model Question Paper			
QP CODE:			
Reg No: _____			
Name: _____			PAGES : 4
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR			
Course Code: 222TCS002			
Course Name: DEEP LEARNING			
Max. Marks : 60		Duration: 2.5 Hours	
PART A			
Answer All Questions. Each Question Carries 5 Marks			
1.		Consider a simple perception f such that $f: \mathbb{R}^4 \rightarrow \mathbb{R}$ which uses sigmoid as its activation function. The inputs are $X=[0.1, 0.2, 0.3, 0.4]$ and the corresponding weights $W=[-0.2, 0.1, 0.2, 0.3]$. Compute the derivative of sigmoid of $z(\sigma'(z))$, where z is a linear combination of weights vector W and input vector X . Ignore the bias term.	
2.		Illustrate convolution and pooling operation with an example.	
3.		Explain the relevance of LSTM in the context of recurrent neural networks. .	
4.		Describe the word2vec technique of word embedding.	
5.		Illustrate the concept of self-attention in transformers.	(5x5=25)
Part B (Answer any five questions. Each question carries 7 marks)			

6.	a	Explain the concepts behind i) Early stopping ii) dropout iii) weight decay	(4)
	b	Using Adagrad-based gradient descent, find the new value of parameter θ_{t+1} , given that the old value $\theta_t=0.7$, aggregated gradient $\Delta\theta_t=0.9$, gradient accumulation $r_{t-1}=0.6$, learning rate $\alpha=0.1$ and small constant $\delta=10^{-8}$.	(3)
7.	a	The input to a CNN architecture is a color image of size $112 \times 112 \times 3$. The first convolution layer comprises 64 kernels of size 5×5 applied with a stride of 2 and padding 0. What will be the number of parameters?	(3)
	b.	Explain how the Vanishing and Exploding gradient problem is addressed in ResNet.	(4)
8.		Explain the working of RNN and discuss how backpropagation through time is used in recurrent networks.	(7)
9.		Illustrate the GloVe technique of word embedding.	(7)
10.		Demonstrate the workflow of the BERT technique of word embedding.	(7)
11.		Justify the need for self-attention in the transformer architecture.	(7)
12.		Illustrate the architecture of a Vision transformer.	(7)

Syllabus

Module 1: Introduction to Deep Learning	(8 hours)
Deep Learning vs traditional machine learning, Gradient Descent, Adam Optimization, Weight initialization strategies, Batch Normalization, Regularization techniques, Cross entropy loss function	
Module 2: Convolutional Neural Networks	(9 hours)
Convolution operation, CNN layers, Building a CNN model, Training a CNN, Deep Autoencoders,	
Module 3: Recurrent Neural Networks	(8 hours)
Basic architecture and variants, Backpropagation through time, LSTM, Deep recurrent neural networks, Machine Translation, Encoder Decoder architecture, Sequence to sequence learning, Beam search	
Module 4: Text processing using deep learning	(8 hours)
Word embeddings, word2vec, GloVe, Subword embedding, BERT	

Module 5: Transformers

:COMPUTER SCIENCE AND ENGINEERING-CS2

(8 hours)

Bahdanau attention, Multi-head attention, Self-attention and positional encoding, Transformer architecture, Transformers for vision, Large-scale pretraining with transformers

Course Plan

No	Topic	No. of Lectures
1	Introduction to Deep Learning	
1.1	Deep Learning vs traditional machine learning	1
1.2	Gradient Descent <i>Eqn, Learning Rate, Adv, Disadv, Batch, Stochastic</i>	1
1.3	Challenges in Neural Network Optimization- Ill conditioning, local Minim, Saddle Points, Exploding Gradients	1
1.4	Adagrad, RMSProp, Adam Optimization <i>5 but box 8p</i>	1
1.5	Weight initialization strategies	1
1.6	Batch Normalization	1
1.7	Regularization - Parameter Norm Penalties L2, L1 Regularization	1
1.8	Regularization - Dataset Augmentation, Noise Robustness, Early Stopping, Dropouts	1
2	Convolutional Neural Networks	
2.1	Convolution operation	1
2.2	Convolution - Sparse Interaction, Parameter sharing, Equi-variant Representation	1
2.3	Pooling, Convolutional Neural Networks, Parameters <i>Max vs Avg (2.5)</i>	1
2.4	Variants of basic convolution function- dilated convolution, Transpose Convolution, 1x1 convolution, 3D Convolution	1
2.5	Backpropagation in Convolutional Layers <i>(2)</i>	1
2.6	Cross Entropy Loss Function, AlexNet <i>(5)</i> CNN-adv <i>(5)</i>	1
2.7	GoogleNet Architecture, Inception Module, ResNet	1
2.8	Autoencoders- sparse autoencoders <i>(5)</i>	1
2.9	Denoising autoencoders	1
3	Recurrent Neural Networks	
3.1	Unfolding Graphs <i>(1) + BPTT (7)</i>	1
3.2	Recurrent Neural Networks, Computing Gradients	1
3.3	Modeling Sequences Conditioned on Context with RNNs	1

types (5)
RNN

RNN vs FNN (5)

3.4	Encoder-Decoder Sequence-to-Sequence Architecture, Applications	1
3.5	Deep recurrent neural networks, Recursive Neural Networks	1
3.6	Exploding and Vanishing Gradients	1
3.7	LSTM	
3.8	Beam search	1
4	Word embedding, BERT	
4.1	Word embeddings	1
4.2	word2vec-Skip-gram and Continuous Bag of words	1
4.3	Word Embedding with Global Vectors (GloVe)	1
4.4	Subword embedding	1
4.5	Hierarchical Softmax	1
4.6	Bidirectional Encoder Representations from Transformers (BERT)	1
4.7	Natural Language Processing: with pretraining using word embedding	1
4.8	Case studies - Sentiment Analysis	1
5	Transformers	
5.1	Bahdanau attention	1
5.2	Multi-head attention	1
5.3	Self-attention and positional encoding	1
5.4	Transformer architecture	1
5.5	Transformers for vision (Lecture 1)	1
5.6	Transformers for vision (Lecture 2)	1
5.7	Large-scale pretraining with transformers (Lecture 1)	1
5.8	Large-scale pretraining with transformers (Lecture 2)	1

Reference Books

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, Second edition, MIT Press, 2016.
2. M. Gopal, Deep Learning, Pearson, 2022
3. Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, Dive into Deep Learning, available online at d2l.ai,